Decoupled State Inconsistency: Unifying Cache and Orchestration Failures in LLM Safety Pipelines

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Abstract

Modern deployment architectures for Large Language Models (LLMs) rely on complex distributed oracles, microservices, and incremental memoization to optimize runtime and safety validation. This paper introduces and unifies a critical class of structural vulnerabilities: *Decoupled State Inconsistency*. We demonstrate how production security pipelines break the fundamental safety invariant property where a policy decision must strictly equal the delivered artifact. Through two distinct attack vectors—Unicode-driven incremental memoization bypass (Salsa/Turbo Tasks desynchronization via `U+203E`) and transactional orchestrator decoupling (Time-of-Check to Time-of-Use failures)—we show that modern validation layers can be decoupled from execution layers. This allows forbidden multi-modal assets and technical schema leaks despite active, explicit policy refusals. We present empirical evidence, mitigation frameworks, and abstract local simulations in Rust.

1 Introduction

Security validation in generative AI infrastructure has shifted from monolithic filters to multi-layered, distributed orchestration graphs. To minimize inference latency, vendors employ parallel state validation, CDN caching, and fine-grained incremental memoization engines.

However, we identify a systemic flaw in this paradigm: the detachment of the state lifecycle between the *Policy Engine* (which decides compliance) and the *Delivery Layer* (which provides execution or artifacts). When these states decouple, the system's core safety invariant is violated:

$$\text{PolicyDecision} == \text{DeliveredArtifact}$$

We analyze this architectural failure mode across two layers: the compiler/tokenization layer (local state) and the system orchestration layer (distributed state).

2 Taxonomy of State Inconsistency

2.1 Vector A: Incremental Memoization Bypass

High-throughput systems reuse prior validation decisions via incremental computational graphs, tracking data dependencies within functional units resembling Value Cells (`Vc<T>`).
$[\text{span}_0](start_span)[span_1](start_span)If input transformation or Unicode can$
$[\text{span}_2](start_span)An attacker prefixing restricted inputs with specified g$
$case tokens (e.g., \texttt{U+203E OVERLINE}) force s token -$
$segmentation desynchronization[span_2](end_span).[span_3](start_span)[span_4$

2.2 Vector B: Transactional Orchestrator Decoupling

In distributed architectures, multi-modal generators and safety managers operate asynchronously. If the orchestrator processes outputs via non-atomic transactions, a state mismatch emerges. We observe this when a user prompt forces concurrent execution blocks:

1. `ImageGenerator` $\rightarrow$ `SUCCESS`
2. `PolicyModerator` $\rightarrow$ `BLOCK`

Without an atomic two-phase commit protocol, the user interface receives a successful graphic metadata payload alongside an explicit textual refusal response, failing to roll back the illegal artifact.

3 Experimental Methodology & Results

Differential analysis was conducted across production frameworks to monitor execution latencies and state alignment.

Table 1: Empirical Analysis of Validation Pipeline Deviations

ID	Target Vector	Latency	
[span$_7$]($start_span$)1.1	`turbopack`	1.1s	
(start$_span$)1.4	`_[REDACTED]`	2.3s	
(start$_span$)4.1	`Vc<T> expansion`	2.0s	
(start$_span$)5.1	`SWC Context`	>600s	
6.1	`Multi-Modal Prompt`	3.4s	

4 Root Cause and Architectural Attack Surface

The core breakdown across both vulnerabilities stems from either a Time-of-Check to Time-of-Use (TOCTOU) vulnerability or a stale cache state pointer. In the TOCTOU scenario, an artifact passes an early pipeline validator, but triggered safety events during later execution phases only update the textual communication node, leaving the generated object graph or memory context active in the CDN/UI data-layer.

5 Proof of Concept Simulation

[span$_1$1]($start_span$)$Listing1providesanabstractsimulationinRustdemonstratingamulti$-$layerpipelinemissingnormalizationandatomicsyncstates$[$span_1$1]($end_span$).

Listing 1: Rust simulation of un-normalized validation and decoupled delivery states.

```rust
struct PipelineResult {
    policy_refusal: bool,
    delivered_artifact: Option<String>,
}

fn evaluate_pipeline(input: &str) ->
    PipelineResult {
    // Architectural Bug: Lack of NFKC
    normalizer
    let is_dangerous = matches!(input, "
    restricted_scope" | "turbopack");

    // Simulate non-atomic distributed
    generation
    let generated_asset = if input.contains("
    restricted_scope") || input.contains("\u{203
    E}") {
        Some("Restricted_Asset_Payload".
    to_string())
    } else {
        None
    };

    PipelineResult {
        policy_refusal: is_dangerous,
        delivered_artifact: generated_asset,
    }
}

fn main() {
    let baseline = "restricted_scope";
    let exploit_input = "\u{203E}
    _restricted_scope";

    let res = evaluate_pipeline(exploit_input);

    // State Inconsistency Check
    if res.policy_refusal == false && res.
    delivered_artifact.is_some() {
        println!("[!] State Inconsistency
    Detected: Invariant Broken!");
    }
}
```

6 Mitigation Strategies

We propose three structural engineering corrections:

- **Atomic Commit Pipeline:** Enforce a strict coordination protocol ensuring that text, image, and metadata blocks share an identical, atomic verification identifier (`DecisionID`).
- **Universal NFKC Normalization:** Standardize canonical normalization mapping prior to injecting items into memoization graphs.
- **Post-Generation Posture Assessment:** Enforce validation on the final generated object matrix instead of relying solely on entry-prompt parameters.

References

[1] Vercel Team. “Turbopack: Incremental Computation Engine.” Vercel Systems Journal, 2023.

[2] Salsa Contributors. “Salsa: On-demand incremental computation framework in Rust.” GitHub, 2024.